张杰

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张杰博士生于 1990 年 10 月 20 日，现任北京大学计算机学院助理教授、特聘研究员、博士生导师，获得优秀青年科学基金海外项目（海外优青）、北京市自然科学基金青年重点项目（原北京杰青）、华为奥林帕斯先锋奖、英特尔学术英才计划荣誉学者、ACM SIGCSE 新星奖、博雅青年学者。他长期从事存储系统和专用处理器的研究和设计，致力于从计算机体系结构层面出发，解决大数据和人工智能时代对于高性能存储系统的需求，突破冯诺依曼体系结构下数据迁移的瓶颈以及内存墙的限制。负责和参与科技部重点研发项目（课题负责人）、国家自然科学基金重点项目（北大负责人）、国家科技重大专项和面上项目，得到华为、江波龙、三星电子、海力士、英特尔和西部数据的资助。自 2016 年起，在计算机国际会议及期刊上发表了 80 余篇论文，包括 ISCA、OSDI、SOSP、HPCA、MICRO、ASPLOS、FAST、ATC、NSDI、DAC 和 Eurosys。**2025 年入选 HPCA 名人堂（首位 90 后成员）。在计算机系统结构领域 90 后学者中，发表顶级会议论文数排名全球第一（CSRankings，2016-2026）。亚洲历史上第三位在所有体系结构和系统顶级会议上发表论文的学者。**论文获 2019 年韩国计算机大会（KCC）最佳演讲论文奖和 2019 年非易失性内存研讨会（NVMW）优秀论文提名奖。在存储领域的研究成果被列为 KAIST50 年重大突破之一，并被四十余家新闻媒体报道。

工作经历

北京大学，北京，中国 — 助理教授，博导
计算机学院 — 2021 年 7 月–至今
KAIST, Daejeon, Korea — 博后
Electrical Engineering (Computing Division) — 2020 年 3 月 –2021 年 5 月

教育经历

Yonsei University, Incheon, Korea — **Advisor:** Dr. Myoungsoo Jung
PhD in Engineering — August 2015 – Feb 2020
University of Texas at Dallas, Richardson, Texas — **Advisor:** Dr. Myoungsoo Jung
PhD in Computer Engineering (transfer to Korea) — August 2014 – August 2015
University of Texas at Dallas, Richardson, Texas — **Advisor:** Dr. Myoungsoo Jung
Master of Science in Electrical Engineering — August 2012 – May 2014
Nanjing University of Posts and Telecommunications, Nanjing, China
BS in Communication Engineering (computer communication) — September 2008 – July 2012

论文发表情况

2027

HPCA — **Twig: Parallel and Memory-Efficient Tree Speculative Decoding for Hybrid-Attention LLMs**
CCF-A 会议
Li Wang, Yi Su, Xiabao Wu, Chiran You, Yongchao Liu, Zhan Qiu, Juelu Zhang, Jiajun Zheng, Fangxin Liu, Jie Zhang, Chen Tian, Chengying Huan
IEEE International Symposium on High-Performance Computer Architecture

TC — **Adaptive Management with Learning-Based Page-Level FTL for Flash-Based SSDs**
CCF-A 期刊
*Shengzhe Wang, Zihang Lin, Suzhen Wu, Hong Jiang, **Jie Zhang**, Bo Mao*
IEEE Transactions on Computers

Eurosys — **Beyond Overlap: Hierarchical Graph Abstraction for Out-of-Core Vector Search**
CCF-A 会议
*Lizheng Chen, Pinhuan Huang, Shaonan Ma, **Jie Zhang**, Qing Wang, Zhengyi Yang, Heng Zhang, Mingxing Zhang, Fan Yang, Guihai Chen, Chen Tian, Chengying Huan*

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22th ACM European Conference on Computer Systems

2026

USENIX ATC

SplineFS: Adaptive Learned Directory Indexing for Memory-Constrained Filesystems

CCF-A 会议

*Jianqiang Zeng, Shaocong Sun, Chengying Huan, Peng Wang, Ke Zhou, Yingwei Luo, Xiaolin Wang, Zhenlin Wang, Diyu Zhou, Yingdi Shan, **Jie Zhang***

ACM SIGOPS Annual Technical Conference

TACO

Cambricon-FlexLLM: A Flexible Chiplet-Based Hybrid Architecture for On-Device 70B LLM Inference

CCF-A 期刊

*Zhongkai Yu, Shengwen Liang, Tianyu Ma, Yunke Cai, Ziyuan Nan, Di Huang, Xinkai Song, Yifan Hao, **Jie Zhang**, Tian Zhi, Yongwei Zhao, Zidong Du, Xing Hu Qi Guo, Tianshi Chen*

Transactions on Architecture and Code Optimizations

MICRO

LUTHE: Breaking the Memory Wall of High-Precision Encrypted LLM Nonlinearities via GPU–CSD Co-Design

CCF-A 会议

*Xiurui Pan, Guoci Chen, Guangyu Sun, Mingzhe Zhang, **Jie Zhang***

The 58th Annual IEEE/ACM International Symposium on Microarchitecture

SC

TPage: Transparent Page Management for CXL-Based Tiered Memory Systems

CCF-A 会议

*Maoxin Ye，Suzhen Wu, **Jie Zhang**, Bo Mao*

International Conference for High Performance Computing, Networking, Storage and Analysis

ICS

Latency-SLO-Aware Memory Offloading for Large Language Model Inference

CCF-B 会议

*Chenxiang Ma, Hanyu Zhao, Zhisheng Ye, Zehua Yang, Tianhao Fu, Jiaxun Han, **Jie Zhang**, Yingwei Luo, Xiaolin Wang, Zhenlin Wang, Yong Li, Diyu Zhou*

ACM International Conference on Supercomputing

HPDC

BCCE: Block-Centric GPU Co-Design for Real-Time Range-Top-K Query at Scale

CCF-A 会议

*Chengying Huan, Ziheng Meng, Zhengyi Yang, Yongchao Liu, **Jie Zhang**, Qing Wang, Jing Wang, Shaonan Ma, Zhibin Wang, Mingxing Zhang, Rong Gu, Baokun Wang, Guihai Chen, Chen Tian*

ACM International Symposium on High-Performance Parallel and Distributed Computing

OSDI

Espresso: Constructing Cost-Efficient CXL JBOF via Inter-SSD Computing Resource Sharing

CCF-A 会议

*Shushu Yi, Yuda An, Li Peng, Xiurui Pan, Qiao Li, Jieming Yin, Guangyan Zhang, Wenfei Wu, Chenxi Wang, Diyu Zhou, Zhenlin Wang, Xiaolin Wang, Yingwei Luo, Ke Zhou, **Jie Zhang***

USENIX Symposium on Operating Systems Design and Implementation

OSDI

Merlin: An Efficient Adaptive Cache Eviction Algorithm via Fine-Grained Characterization

CCF-A 会议

*Liujia Li, Jinhao Guo, Yi Fan, Jianyu Wu, Zhenling Wang, **Jie Zhang**, Yuval Tamir, Xiaolin Wang, Yingwei Luo, Diyu Zhou*

USENIX Symposium on Operating Systems Design and Implementation

OSDI

MultiLane: Eliminating Centralized Bottlenecks in User-space Network Stack

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CCF-A 会议 *Kang Hu, Shuqi Dong, Chuandong Li, Ran Yi, Zonghao Zhang, **Jie Zhang**, Yingwei Luo, Xiaolin Wang, Zhenling Wang, Diyu Zhou*

USENIX Symposium on Operating Systems Design and Implementation

ICLR **LouisKV: Efficient KV Cache Retrieval for Long Input-Output Sequences**

CCF-A 会议 *Wenbo Wu, Qingyi Si, Xiurui Pan, Ye Wang, **Jie Zhang****

The Fourteenth International Conference on Learning Representations

DAC **WILL: Write Invalidation Skipping for QLC NAND by Leveraging Data Lifespan and Latency Awareness**

CCF-A 会议 *Beichen Ning, Yina Lv, Tianyu Ren, Bin Gao, Xiangyu Yao, **Jie Zhang**, Qiao Li and Chun Jason Xue*

Design Automation Conference

DAC **Cachence: Fine-Grained Cache Partitioning in Both Time and Space**

CCF-A 会议 *Liujia Li, Yuanlong Li, Yiming Yao, Jianyu Wu, Yi Fan, Jinhao Guo, Liren Zhu, **Jie Zhang**, Xiaolin Wang, Yingwei Luo, Zhenlin Wang, Diyu Zhou*

Design Automation Conference

FAST **Xerxes: Extensive Exploration of Scalable Hardware Systems with CXL-Based Simulation**

CCF-A 会议 **Framework**

*Yuda An, Shushu Yi, Bo Mao, Qiao Li, Mingzhe Zhang, Diyu Zhou, Ke Zhou, Nong Xiao, Guangyu Sun, Yingwei Luo, **Jie Zhang****

The USENIX Conference on File and Storage Technologies

HPCA **AutoGNN: End-to-End Hardware-Driven Graph Preprocessing for Enhanced GNN**

CCF-A 会议 **Performance**

*Seungkwan Kang, Seungjun Lee, Donghyun Gouk, Miryeong Kwon, Hyunkyu Choi, Junhyeok Jang, Sangwon Lee, Huiwon Choi, **Jie Zhang**, Wonil Choi, Mahmut T Kandemir, Myoungsoo Jung*

IEEE International Symposium on High-Performance Computer Architecture

Eurosys **ColdCode: Cold Data Encoding for Enhanced Reliability and Lifetime in 3D NAND Flash**

CCF-A 会议 *Qiao Li, Shangyu Wu, Zheng Wan, Yufei Cui, **Jie Zhang****, *Chun Jason Xue*

21th ACM European Conference on Computer Systems

2025

TCAD **CXL-DMSim: A Full-System CXL Disaggregated Memory Simulator With Comprehensive Silicon Validation**

CCF-A 期刊 *Yanjing Wang, Lizhou Wu, Wentao Hong, Yang Ou, Zicong Wang, Sunfeng Gao, **Jie Zhang**, Sheng Ma, Dezun Dong, Xingyun Qi, Mingche Lai, Nong Xiao*

ACM Transactions on Computer-Aided Design of Integrated Circuits and Systems

PACT **Exploring Memory Tiering Systems in the CXL Era via FPGA-based Emulation and Device-**

CCF-B 会议 **Side Management**

*Yiqi Chen, Xiping Dong, Zhe Zhou, Zhao Wang, **Jie Zhang**, Guangyu Sun,*

International Conference on Parallel Architecture and Compilation Techniques

NEWTON **Ultrafast and Reliable Domain-Wall and Skyrmion Logic in a Chirally Coupled Ferrimagnet**

CELL 子刊 *Yifei Ma, Dihua Wu, Fengbo Yan, Xiaoxiao Fang, Peixin Qin, Leran Wang, Li Liu, Laichuan Shen, Zhiqi Liu, Wenyun Yang,* ***Jie Zhang****, Yanglong Hou, Yan Zhou, Feng Luo, Jinbo Yang, Kaiming Cai, Shuai Ning, Zhaochu Luo*

Cell Press NEWTON

SOSP **Aeolia: Fast and Secure Userspace Interrupt-Based Storage Stack**

CCF-A 会议 *Chuandong Li, Ran Yi, Zonghao Zhang, Jing Liu, Changwoo Min,* ***Jie Zhang****, Yingwei Luo, Xiaolin Wang, Zhenling Wang, Diyu Zhou*

ACM Symposium on Operating Systems Principles

HotStorage **SPDK+: Low Latency or High Power Efficiency? We Take Both**

Endian Li, Shushu Yi, Li Peng, Qiao Li, Diyu Zhou, Zhenlin Wang, Xiaolin Wang, Bo Mao,

Yingwei Luo, Ke Zhou, ***Jie Zhang****

The 17th ACM Workshop on Hot Topics in Storage and File Systems

ICDE **BL-Tree: The Best of Both Worlds by combining B+-Tree on Top and LSM-Tree on Bottom**

CCF-A 会议 *Maoxin Ye, Zuocheng Wang, Shengzhe Wang, Jiahong Chen, Chunfeng Du,* ***Jie Zhang****, Ke Zhou, Suzhen Wu, Bo Mao*

IEEE International Conference on Data Engineering

ToS **Enhancing the Performance of Next-Generation SSD Arrays: A Holistic Approach**

CCF-A 期刊 ***Jie Zhang****, Shushu Yi, Xiurui Pan, Qiao Li, Qiang Li, Chenxi Wang, Bo Mao, Myoungsoo Jung*

ACM Transactions on Storage

TCAD **FlexHMB: A Flexible HMB Design towards Bufferless Mobile Flash**

CCF-A 期刊 *Yong Peng, Li Peng, Shaocong Sun, Lujia Yin, Qiao Li,* ***Jie Zhang****

ACM Transactions on Computer-Aided Design of Integrated Circuits and Systems

ISCA **XHarvest: Rethinking High-Performance and Cost-Efficient SSD Architecture with**

CCF-A 会议 **CXL-Driven Harvesting**

Li Peng, Wenbo Wu, Shushu Yi, Xianzhang Chen, Chenxi Wang, Shengwen Liang, Zhe Wang, Nong Xiao, Qiao Li, Mingzhe Zhang, ***Jie Zhang****

IEEE/ACM International Symposium on Computer Architecture

ISCA **ArtMem: Adaptive Migration in Reinforcement Learning-Enabled Tiered Memory**

CCF-A 会议 *Xinyue Yi, Hongchao Du, Yu Wang,* ***Jie Zhang****, Qiao Li, Chun Jason Xue*

IEEE/ACM International Symposium on Computer Architecture

Eurosys **Daredevil: Rescue Your Flash Storage from Inflexible Kernel Storage Stack**

CCF-A 会议 *Junzhe Li, Ran Shu, Jiayi Lin, Qingyu Zhang, Ziyue Yang,* ***Jie Zhang****, Yongqiang Xiong, Chenxiong Qian*

20th ACM European Conference on Computer Systems

NSDI **Beehive: A Scalable Disaggregated Memory Runtime Exploiting Asynchrony of**

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CCF-A 会议	**Multithreaded Programs**
	*Quanxi Li, Hong Huang, Ying Liu, Yanwen Xia, **Jie Zhang**, Mosong Zhou, Xiaobing Feng, Huimin Cui, Quan Chen, Yizhou Shan, Chenxi Wang*
	22nd USENIX Symposium on Networked Systems Design and Implementation
HPCA	**InstAttention: In-Storage Attention Offloading for Cost-Effective Long-Context LLM**
CCF-A 会议	**Inference**
	*Xiurui Pan, Endian Li, Qiao Li, Shengwen Liang, Yizhou Shan, Ke Zhou, Yingwei Luo, Xiaolin Wang, **Jie Zhang***
	IEEE International Symposium on High-Performance Computer Architecture
HPCA	**NeuVSA: A Unified and Efficient Accelerator for Neural Vector Search**
CCF-A 会议	*Ziming Yuan, Lei Dai, Wen Li, **Jie Zhang**, Shengwen Liang, Ying Wang, Cheng Liu, Huawei Li, Jiafeng Guo, Peng Wang, Renhai Chen, Gong Zhang*
	IEEE International Symposium on High-Performance Computer Architecture
HPCA	**Criticality-Aware Instruction-Centric Bandwidth Partitioning for Data Center Applications**
CCF-A 会议	*Liren Zhu, Liujia Li, Jianyu Wu, Yiming Yao, Zhan Shi, **Jie Zhang**, Zhenlin Wang, Xiaolin Wang, Yingwei Luo, Diyu Zhou*
	IEEE International Symposium on High-Performance Computer Architecture
	2024
SOSP	**BIZA: Design of Self-Governing Block-Interface ZNS AFA for Endurance and Performance**
CCF-A 会议	*Shushu Yi, Shaocong Sun, Li Peng, Yingbo Sun, Ming-Chang Yang, Zhichao Cao, Qiao Li, Myoungsoo Jung, Ke Zhou, **Jie Zhang***
	ACM Symposium on Operating Systems Principles
MICRO	**NeoMem: Hardware/Software Co-Design for CXL-Native Memory Tiering**
CCF-A 会议	*Zhe Zhou, Yiqi Chen, Tao Zhang, Yang Wang, Ran Shu, Shuotao Xu, Peng Cheng, Lei Qu, Yongqiang Xiong, **Jie Zhang**, Guangyu Sun*
	The 57th Annual IEEE/ACM International Symposium on Microarchitecture
MICRO	**Cambricon-llm: A chiplet-based hybrid architecture for on-device Inference of 70B LLM**
CCF-A 会议	*Zhongkai Yu, Shengwen Liang, Tianyu Ma, Yunke Cai, Ziyuan Nan, Di Huang, Xinkai Song, Yifan Hao, **Jie Zhang**, Tian Zhi, Yongwei Zhao, Zidong Du, Xing Hu Qi Guo, Tianshi Chen*
	The 57th Annual IEEE/ACM International Symposium on Microarchitecture
ToS	**Extremely-Compressed SSDs with I/O Behavior Prediction**
CCF-A 期刊	*Xiangyu Yao, Qiao Li, Kaihuan Lin, Xinbiao Gan, **Jie Zhang**, Congming Gao, Zhirong Shen, Quanqing Xu, Chuanhui Yang, Jason Xue*
	ACM Transactions on Storage
TC	**Land of Oz: Resolving Orderless Writes in Zoned Namespace SSDs**
CCF-A 期刊	*Yingjia Wang, You Zhou, Fei Wu, **Jie Zhang**, Ming-Chang Yang*
	IEEE Transactions on Computers

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USENIX ATC	**ScalaAFA: Constructing User-Space All-Flash Array Engine with Holistic Designs**
CCF-A 会议	*Shushu Yi, Xiurui Pan, Qiao Li, Qiang Li, Chenxi Wang, Bo Mao, Myoungsoo Jung,* ***Jie Zhang****
	USENIX Annual Technical Conference
USENIX ATC	**ScalaCache: Scalable User-Space Page Cache Management with Software-Hardware**
CCF-A 会议	**Coordination**
	Li Peng, Yuda An, You Zhou, Chenxi Wang, Qiao Li, Chuanning Chen, ***Jie Zhang****
	USENIX Annual Technical Conference
Arxiv	**From RDMA to RDCA: Toward High-Speed Last Mile of Data Center Networks Using Remote Direct Cache Access**
	Qiang Li, Qiao Xiang, Derui Liu, Yuxin Wang, Haonan Qiu, Xiaoliang Wang, ***Jie Zhang****, Ridi Wen, Haohao Song, Gexiao Tian, Chenyang Huang, Lulu Chen, Shaozong Liu, Yaohui Wu, Zhiwu Wu, Zicheng Luo, Yuchao Shao, Chao Han, Zongjie WU, Jianbo Dong, Zheng Cao, Jinbo WU, Jiwu Shu, Jiesheng Wu*
	Arxiv
TACO	**Characterizing and Optimizing LDPC Performance on 3D NAND Flash Memories**
CCF-A 期刊	*Qiao Li, Yu Chen, Guanyu Wu, Yajuan Du, Min Ye, Xinbiao Gan,* ***Jie Zhang****, Zhirong Shen, Jiwu Shu, Chun Jason Xue*
	ACM Transactions on Architecture and Code Optimization
ISCA	**Flagger: Cooperative Acceleration for Large-Scale Cross-Silo Federated Learning**
CCF-A 会议	**Aggregation**
	Xiurui Pan, Yuda An, Shengwen Liang, Bo Mao, Mingzhe Zhang, Qiao Li, Myoungsoo Jung, ***Jie Zhang****
	IEEE/ACM International Symposium on Computer Architecture
ASPLOS	**Achieving Near-Zero Read Retry for 3D NAND Flash Memory**
CCF-A 会议	*Min Ye, Qiao Li, Yina Lv,* ***Jie Zhang****, Tianyu Ren, Daniel Wen, Tei-Wei Kuo, Chun Jason Xue*
	ACM International Conference on Architectural Support for Programming Languages and Operating Systems
HPCA	**StreamPIM: Streaming Matrix Computation in Racetrack Memory**
CCF-A 会议	*Yuda An, Yunxiao Tang, Shushu Yi, Li Peng, Xiurui Pan, Guangyu Sun, Zhaochu Luo, Qiao Li,* ***Jie Zhang****
	IEEE International Symposium on High-Performance Computer Architecture
HPCA	**BeaconGNN: Large-Scale GNN Acceleration with Asynchronous In-Storage Computing**
CCF-A 会议	*Yuyue Wang, Xiurui Pan, Yuda An,* ***Jie Zhang****, Glenn Reinman**
	IEEE International Symposium on High-Performance Computer Architecture
HPCA	**LearnedFTL: A Learning-based Page-level FTL for Reducing Double Reads in Flash-based SSDs**
CCF-A 会议	*Shengzhe Wang, Zihang Lin, Suzhen Wu, Hong Jiang,* ***Jie Zhang****, Bo Mao*
	IEEE International Symposium on High-Performance Computer Architecture

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HPCA **Midas Touch: Invalid-Data Assisted Reliability and Performance Boost for 3D High-Density Flash**

CCF-A 会议 *QiaoLi, Hongyang Dang, Zheng Wan, Congming Gao, Min Ye, **Jie Zhang**, Tei-Wei Kuo, Chun Jason Xue*

IEEE International Symposium on High-Performance Computer Architecture

2023

NVMW **Optimizations of Linux Software RAID System for Next-Generation Storage**

*Shushu Yi, Yanning Yang, Yunxiao Tang, Zixuan Zhou, Junzhe Li, Yue Chen, Myoungsoo Jung, **Jie Zhang***

The 14th Annual Non-volatile Memories Workshop

SAC **BcBench: Exploring Throughput Processor Designs based on Blockchain Benchmarking**

*Xiurui Pan, Yue Chen, Shushu Yi, **Jie Zhang***

The 38th ACM/SIGAPP Symposium on Applied Computing

CAL **Intelligent SSD Firmware for Zero-Overhead Journaling**

SCI 3 区 *Hanyeoreum Bae, Donghyun Gouk, Seungjun Lee, Jiseon Kim, Sungjoon Koh, **Jie Zhang***, Myoungsoo Jung*,*

IEEE Computer Architecture Letters (CAL)

2022

HotStorage **ScalaRAID: Optimizing Linux Software RAID System for Next-Generation Storage**

*Shushu Yi, Yanning Yang, Yunxiao Tang, Zixuan Zhou, Junzhe Li, Yue Chen, Myoungsoo Jung, **Jie Zhang***

14th USENIX Workshop on Hot Topics in Storage and File Systems (HotStorage 22)

THPC **Survey on Storage-Accelerator Data Movement**

*Zixuan Zhou, Shushu Yi, **Jie Zhang***

CCF Transaction on High Performance Computing

NVMW **Integrating New Photonic-Based Heterogeneous Memory into Throughput Accelerators**

***Jie Zhang**, Myoungsoo Jung,*

The 13th Annual Non-volatile Memories Workshop

NVMW **HAMS: Hardware Automated Memory-over-Storage for Large-scale Memory Expansion**

***Jie Zhang**, Miryeong Kwon, Donghyun Gouk, Sungjoon Koh, Nam Sung Kim,*

Mahmut Taylan Kandemir, Myoungsoo Jung

The 13th Annual Non-volatile Memories Workshop

2021

MICRO **Ohm-GPU: Integrating New Optical Network and Heterogeneous Memory into GPU**

CCF-A 会议 **Multi-Processors**

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The 54th Annual IEEE/ACM International Symposium on Microarchitecture

ISCA **Revamping Storage Class Memory with Hardware Automated Memory-Over-Storage Solution**

CCF-A 会议

***Jie Zhang**, Miryeong Kwon, Donghyun Gouk, Sungjoon Koh, Nam Sung Kim, Mahmut Kandemir, Myoungsoo Jung,*

The IEEE/ACM International Symposium on Computer Architecture

NVMW **Architecting Throughput Processors with New Flash**

***Jie Zhang**, Myoungsoo Jung,*

The 12th Annual Non-volatile Memories Workshop

NVMW **DRAM-less Accelerator for Energy Efficient Data Processing**

***Jie Zhang**, Gyuyoung Park, David Donofrio, John Shalf, Myoungsoo Jung*

The 12th Annual Non-volatile Memories Workshop

NVMW **Manycore-Based Scalable SSD Architecture Towards One and More Million IOPS**

***Jie Zhang**, Miryeong Kwon, Michael Swift, Myoungsoo Jung,*

The 12th Annual Non-volatile Memories Workshop

2020

ISCA **ZnG: Architecting GPU Multi-Processors with New Flash for Scalable Data Analysis**

CCF-A 会议

***Jie Zhang**, Myoungsoo Jung,*

The IEEE/ACM International Symposium on Computer Architecture

FAST **Scalable Parallel Flash Firmware for Many-core Architectures**

CCF-A 会议

***Jie Zhang**, Miryeong Kwon, Michael Swift, Myoungsoo Jung,*

The 18th USENIX Conference on File and Storage Technologies

HPCA **DRAM-less: Hardware Acceleration of Data Processing with New Memory**

CCF-A 会议

***Jie Zhang**, Gyuyoung Park, David Donofrio, John Shalf, Myoungsoo Jung*

26th IEEE International Symposium on High-Performance Computer Architecture

ISPASS **Data Direct I/O Characterization for Future I/O System Exploration**

CCF-C 会议

*Mohammad Alian, Yifan Yuan, **Jie Zhang**, Ren Wang, Myoungsoo Jung, Nam Sung Kim*

The IEEE International Symposium on Performance Analysis of Systems and Software

CAL **FastDrain: Removing Page Victimization Overheads in NVMe Storage Stack**

SCI-3 区

***Jie Zhang**, Miryeong Kwon, Sanghyun Han, Nam Sung Kim, Mahmut Kandemir and Myoungsoo Jung*

IEEE Computer Architecture Letters (CAL)

2019

HPCA **FUSE: Fusing STT-MRAM into GPUs to Alleviate Off-Chip Memory Access Overheads**

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CCF-A 会议 ***Jie Zhang***, *Myoungsoo Jung, Mahmut Kandemir,*
25th IEEE International Symposium on High-Performance Computer Architecture

IISWC **Faster than Flash: An In-Depth Study of System Challenges for Emerging Ultra-Low Latency SSDs**
Sungjoon Koh, Junkyeok Jang, Changrim Lee, Miryeong Kwon, ***Jie Zhang***, *Myoungsoo Jung,*
The 2019 IEEE International Symposium on Workload Characterization

DAC **FlashGPU: Placing New Flash Next to GPU Cores**
CCF-A 会议 ***Jie Zhang***, *Miryeong Kwon, Hyojong Kim, Hyesoon Kim, Myoungsoo Jung,*
The 56th Design Automation Conference (DAC)

TPDS **Exploring Fault-Tolerant Erasure Codes for Scalable All-Flash Array Clusters**
CCF-A 期刊 *Sungjoon Koh,* ***Jie Zhang***, *Miryeong Kwon, Jungyeon Yoon, David Donofrio, Nam Sung Kim, Myoungsoo Jung,*
IEEE Transactions on Parallel and Distributed Systems (TPDS)

NVMW **Addressing Fast-Detrapping for Reliable 3D NAND Flash Design**
Mustafa Shihab, ***Jie Zhang***, *Myoungsoo Jung, Mahmut Kandemir,*
10th Annual Non-Volatile Memories Workshop -- Nominated as Memorable Paper Award

KCC **Maximizing GPU Cache Utilization with Adjustable Cache Line Management**
Jie Zhang, *Myoungsoo Jung,*
Korean Computer Congress (KCC), 2019 -- Nominated as Excellent Paper Award

2018

OSDI **FlashShare: Punching Through Server Storage Stack from Kernel to Firmware for**
CCF-A 会议 **Ultra-Low Latency SSDs**
Jie Zhang, *Miryeong Kwon, Donghyun Gouk, Changlim Lee, Mohammad Alian, Myoungjun Chun, Mahmut Kandemir, Nam Sung Kim, Jihong Kim, Myoungsoo Jung,*
13th USENIX Symposium on Operating Systems Design and Implementation

MICRO **Amber: Enabling Precise Full-System Simulation with Detailed Modeling of All SSD**
CCF-A 会议 **Resources**
Donghyun Gouk, Miryeong Kwon, ***Jie Zhang***, *Sungjoon Koh, Wonil Choi, Nam Sung Kim, Mahmut Kandemir, Myoungsoo Jung,*
The 51st Annual IEEE/ACM International Symposium on Microarchitecture

TACO **ReveNAND: A Fast-Drift Aware Resilient 3D NAND Flash Design**
CCF-B 期刊 *Mustafa Shihab,* ***Jie Zhang***, *Myoungsoo Jung, Mahmut Kandemir,*
ACM Transactions on Architecture and Code Optimization (TACO), 2018

Eurosys **FlashAbacus: A Self-governing Flash-based Accelerator for Low-power Systems**

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CCF-A 会议 ***Jie Zhang***, *Myoungsoo Jung,*
The European Conference on Computer Systems (EuroSys), 2018

IPDPS **CIAO: Cache Interference-Aware Throughput-Oriented Architecture and Scheduling for GPUs**
CCF-B 会议 ***Jie Zhang***, *Shuwen Gao, Nam Sung Kim, Myoungsoo Jung,*
32nd IEEE International Parallel & Distributed Processing Symposium (IPDPS), 2018

2017

CAL **SimpleSSD: Modeling Solid State Drive for Holistic System Simulation**
SCI-3 区 *Myoungsoo Jung,* ***Jie Zhang***, *Ahmed Abulila, Miryeong Kwon, Narges Shahidi, John Shalf, Nam Sung Kim and Mahmut Kandemir,*
IEEE Computer Architecture Letters (CAL), 2017

IISWC **Understanding System Characteristics of Online Erasure Coding on Scalable, Distributed and Large-Scale SSD Array Systems**
Sungjoon Koh, ***Jie Zhang***, *Miryeong Kwon, Jungyeon Yoon, David Donofrio, Nam Sung Kim, Myoungsoo Jung,*
IEEE International Symposium on Workload Characterization (IISWC), 2017

IISWC **TraceTracker: Hardware/Software Co-Evaluation for Large-Scale I/O Workload Reconstruction**
Miryeong Kwon, ***Jie Zhang***, *Gyuyoung Park, Wonil Choi, David Donofrio, John Shalf, Mahmut Kandemir, Myoungsoo Jung,*
IEEE International Symposium on Workload Characterization (IISWC), 2017

NPC **An In-depth Performance Analysis of Many-Integrated Core for Communication Efficient Heterogeneous Computing**
CCF-C 会议 ***Jie Zhang***, *Myoungsoo Jung,*
IFIP International Conference on Network and Parallel Computing (NPC), 2017

NPC/IJPP **Enabling Realistic Logical Device Interface and Driver for NVM Express Enabled Full System Simulations**
CCF-C 会议 *Donghyun Gouk,* ***Jie Zhang***, *Myoungsoo Jung,*
IFIP International Conference on Network and Parallel Computing (NPC) and Invited for International Journal of Parallel Programming (IJPP), 2017

2016

HPCA **DUANG: Fast and Lightweight Page Migration in Asymmetric Memory Systems**
CCF-A 会议 *Hao Wang,* ***Jie Zhang***, *Gieseo Park, Sharmila Shridhar, Myoungsoo Jung, Nam Sung Kim,*
IEEE Symposium on High Performance Computer Architecture (HPCA), 2016

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ASBD **A Study for Block-level I/O Trace Reconstruction on All-Flash Arrays**
*Miryeong Kwon, **Jie Zhang**, Gyuyoung Park, Myoungsoo Jung,*
Workshop on Architectures and Systems for Big Data (ASBD@ISCA), 2016

NVMSA **An In-Depth Study of Next Generation Interface for Emerging Non-Volatile Memories**
*Wonil Choi, **Jie Zhang**, Shuwen Gao, Jaesoo Lee, Myoungsoo Jung, Mahmut Kandemir,*
IEEE Non-Volatile Memory Systems and Applications Symposium (NVMSA), 2016

INFLOW **ROSS: A Design of Read-Oriented STT-MRAM Storage for Energy-Efficient Non-Uniform Cache Architecture**
Jie Zhang, Miryeong Kwon, Chanyoung Park, Myoungsoo Jung, Songkuk Kim,
USENIX Workshop on Interactions of NVM/Flash with Operating Systems and Workloads

INFLOW **Couture: Tailoring STT-MRAM for Persistent Main Memory**
*Mustafa Shihab, **Jie Zhang**, Shuwen Gao, Josep Sloan, Myoungsoo Jung,*
USENIX Workshop on Interactions of NVM/Flash with Operating Systems and Workloads

2015

ASBD **CoDEN: A Hardware/Software CoDesign Emulation Platform for SSD-Accelerated Near Data Processing**
***Jie Zhang**, Damian Szmulewicz, Erick Macias, Myoungsoo Jung,*
The 5th Workshop on Architecture and System for Big Data (ASBD), 2015

PACT **NVMMU: Direct Solid State Disk Access for GPU-Accelerated Data Processing**
CCF-B 会议 ***Jie Zhang**, David Donofrio, John Shalf, Myoungsoo Jung,*
The 24th International Conference on Parallel Architecture and Compilation Techniques

ICCD **OpenNVM: An Open-Sourced FPGA-based NVM Controller for Low Level Memory Characterization**
CCF-B 会议 ***Jie Zhang**, Gieseo Park, David Donofrio, Mustafa Shihab, John Shalf and Myoungsoo Jung,*
The 33rd International Conference on Computer Design (ICCD), 2015

PACT-SRC **Integrating 3D Resisteive Memory Cache into GPGPU for Energy-Efficient Data Processing**
***Jie Zhang**, David Donofrio, John Shalf and Myoungsoo Jung,*
International Conference on parallel Architecture and Compilation Techniques (PACT) – ACM SRC 2nd Runner Award, 2015

FAST-WiP **Shared Non-Volatile Mmeory Cache for Energy-Efficient High Throughput GPU Computing**
***Jie Zhang** and Myoungsoo Jung,*
USENIX Conference on File and Storage Technologies Working in Progress (FAST WiP), 2015

2014

HotStorage **Power, Energy, and Thermal Considerations in SSD-Based I/O Acceleration**

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***Jie Zhang**, Myoungsoo Jung,*

6th USENIX Workshop on Hot Topics in Storage and File Systems (HotStorage 14), 2014

专利发表情况

- “一种基于原语级强度感知分解的 TFHE 全同态加密高精度非线性求值方法”， 张杰，潘修睿，陈国赐，孙广宇，张明喆 (披露)
- “基于自适应子树学习型索引的文件系统目录查找算法”， 张杰，曾建强，周可，罗英伟 (披露)
- “一种基于用户中断与频率调节和核心缩放协同优化的 NVMe-oF 存储系统功耗控制方法”， 张杰，李恩典，汪正洋 (披露)
- “GPU 原生的高性能全闪阵列软硬件设计”，易舒舒，张杰 (披露)
- “一种面向大模型推理的旁路山层转换层的计算型存储设备及数据处理方法”，张杰，潘修睿，周可 (披露)
- “一种基于主机端资源借用的成本效益型固态硬盘及主机”，张杰，彭力，武文博，易舒舒，陈咸彰，王晨曦，梁胜文，王喆，李乔，张明喆 (披露)
- “基于硬件一致性互联实现盘间资源共享的全闪阵列”，易舒舒，张杰，安昱达，彭力，李乔，尹捷明，吴文斐，周迪宇，王振林，汪小林，罗英伟，周可 (披露)
- “基于软硬协同设计的全闪阵列”，张杰，易舒舒，潘修睿，徐一茗，李乔，李强，王晨曦，毛波，Myoungsoo Jung (披露)
- “Memory controlling device and computing device including the same”, Myoungsoo Jung, Donghyun Gouk, Miryeong Kwon, Sungjoon Koh, Jie Zhang, America (US20190171566A1)
- “Flash-based accelerator and computing device including the same”, Myoungsoo Jung, Jie Zhang, America (US10824341B2, US20180321859, US20170285968)
- “基于闪存的加速器和包含其的计算设备”, Myoungsoo Jung, Jie Zhang, China (CN107291424)
- “基于闪存的加速器及包括该加速器的计算设备”, Myoungsoo Jung, Jie Zhang, China (CN109460369)
- “Resistance switching memory-based accelerator”, Myoungsoo Jung, Gyuyoung PARK, Jie Zhang, America (US20180321880A1)
- “PARALLEL PROCESSING UNIT, COMPUTING DEVICE INCLUDING THE SAME, AND THREAD SCHEDULING METHOD”, Jie Zhang, Myoungsoo Jung, America (WO2018021620)
- “MEMORY CONTROL APPARATUS AND COMPUTING DEVICE INCLUDING SAME”, JUNG MYOUNGSOO, GOUK DONGHYUN, KWON MIRYEONG, KOH SUNGJOON, 정명수, JIE ZHANG, 국동현, 권미령, 고성준 장지에, Korea (KR1020180126267)
- “COMPUTING DEVICE, METHOD OF PROCESSING INPUT/OUTPUT REQUEST, AND RECORDING MEDIUM”, Jie Zhang, Myoungsoo Jung, Donghyun Gouk, Miryeong Kwon, Sungjoon Koh, America (pending)
- “FLASH-BASED COPROCESSOR”, Jie Zhang, Myoungsoo Jung, America (pending)
- “FLASH STORAGE DEVICE AND METHOD OF SCHEDULING PAGE VICTIMIZATION”, Jie Zhang, Myoungsoo Jung, America (pending)

学术活动情况

Journal Guest Editor

- IEEE MICRO Special Issue

Journal Paper Reviewer

- IEEE Transactions on Computer
- ACM Transactions on Storage
- ACM Transactions on Architecture and Code Optimization

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- ACM Transactions on Computer Systems
- IEEE Transactions on Parallel and Distributed Systems
- IEEE Computer Architecture Letters
- IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems

Conference Program Committee Member

- Eurosys'27
- APSys'26
- OSDI'26
- MICRO'25'24
- ISCA'26'25'24
- HPCA'27'26'25'24
- ATC'25'23
- ISPASS'26'25
- USENIX ATC'23
- ChinaSys'24'22
- SAC'22

演讲情况

- Invited talk, "Storage Architecture Design in AI Era", Nanjing University of Posts and Telecommunications, 2026
- Invited talk, "Progress in Near-Data Processing for AI", CCF SYS, 2025
- Invited talk, "Storage Architecture Design in AI Era", CCF China Storage Conference, 2025
- Invited talk, "Memory System Design based on Coherence Interconnects", HPC China, 2025
- Invited talk, "Memory System Design based on Coherence Interconnects", Compunet, 2025
- Invited talk, "Flagger: Cooperative Acceleration for Large-Scale Cross-Silo Federated Learning Aggregation", CNCC, 2024
- Invited talk, "From Emerging Applications to New Storage Technology: A Full-Stack Exploration", Young PI forum, 2024
- Invited talk, "Progress in Near-Data Processing", CCF Computility, 2024
- Invited talk, "Progress in Near-Data Processing", ASSIST, 2024
- Invited talk, "Progress in Near-Data Processing", CCF-TCIST, 2024
- Invited talk, "Progress in Near-Data Processing", Renmin University, 2023
- Invited talk, "Storage System: From Firmware to Architecture", Xiamen University, 2022
- Invited talk, "Revisiting the Designs of Storage Class Memory", CCF CHIP, 2022
- Invited talk, "ZnG: Architecting GPU Multi-Processors with New Flash for Scalable Data Analysis", Intel Computational Storage Lab, 2020
- Presentation, "ZnG: Architecting GPU Multi-Processors with New Flash for Scalable Data Analysis", ISCA, online, 2020
- Presentation, "DRAM-less: Hardware Acceleration of Data Processing with New Memory", HPCA, San Diego, CA, 2020
- Presentation, "Scalable Parallel Flash Firmware for Many-core Architectures", FAST, Santa Clara, CA, 2020
- Presentation, "FUSE: Fusing STT-MRAM into GPUs to Alleviate Off-Chip Memory Access Overheads", HPCA, Washington DC, 2019
- Presentation, "FlashGPU: Placing New Flash Next to GPU Cores", DAC, Las Vegas, NV, 2019

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- Presentation, “Maximizing GPU Cache Utilization with Adjustable Cache Line Management”, Jeju, South Korea, 2019
- Presentation, “FlashShare: Punching Through Server Storage Stack from Kernel to Firmware for Ultra-Low Latency SSDs”, OSDI, Carlsbad, CA, 2018
- Presentation, “FlashAbacus: A Self-governing Flash-based Accelerator for Low-power Systems”, Eurosys, Porto, Portugal, 2018
- Presentation, “CIAO: Cache Interference-Aware Throughput-Oriented Architecture and Scheduling for GPUs”, IPDPS, Vancouver, Canada, 2018
- Presentation, “An In-depth Performance Analysis of Many-Integrated Core for Communication Efficient Heterogeneous Computing”, NPC, Anhui, China, 2017
- Presentation, “ROSS: A Design of Read-Oriented STT-MRAM Storage for Energy-Efficient Non-Uniform Cache Architecture”, Inflow, Savannah, GA, 2016
- Presentation, “Couture: Tailoring STT-MRAM for Persistent Main Memory”, Inflow, Savannah, GA, 2016
- Presentation, “CoDEN: A Hardware/Software CoDesign Emulation Platform for SSD-Accelerated Near Data Processing”, ASBD, Portland, OR, 2015
- Presentation, “NVMMU: Direct Solid State Disk Access for GPU-Accelerated Data Processing”, PACT, San Francisco, CA, 2015
- Presentation, “Integrating 3D Resistive Memory Cache into GPGPU for Energy-Efficient Data Processing”, PACT SRC, San Francisco, CA, 2015
- Presentation, “OpenNVM: An Open-Sourced FPGA-based NVM Controller for Low Level Memory Characterization”, ICCD, New York city, NY, 2015
- Presentation, “Shared Non-Volatile Memory Cache for Energy-Efficient High Throughput GPU Computing”, FAST WiP, Santa Clara, CA, 2015
- Presentation, “Power, Energy, and Thermal Considerations in SSD-Based I/O Acceleration”, HotStorage, Philadelphia, PA, 2014

教学经历

- Advanced Computer Architecture (Fall’26, Fall’25, Fall’24, Fall’23)
- Computer Architecture (Fall’26, Fall’25, Fall’24, Fall’23, Fall’22)
- Introduction of Computer System (Fall’26, Fall’25, Fall’24, Fall’23, Fall’22)
- IIT 3002 Operating Systems (Fall’15, Fall’16)
- IIT 6036 Computer Organization and Design (Fall’15, Fall’16)
- IIT 7024 Advanced System Architecture (Spring’16)

获奖情况

- 2026：北京市自然科学基金青年重点项目（原北京杰青）
- 2025：HPCA 名人堂
- 2025：奥林帕斯先锋奖
- 2023: 英特尔学术英才计划荣誉学者
- 2023: ACM SIGCSE 新星奖
- 2022: 国家自然科学基金优秀青年基金海外项目（海外优青）
- 2021: Our storage-class memory research is selected as KAIST breakthrough 50 years
- 2020-2021: Korean BK21+ Scholarship
- 2020: HPCA travel grant

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- 2019: Annual Non-Volatile Memories Workshop (NVMW) -- Nominated as Memorable Paper Award
- 2019: Korea Computer Congress (KCC) -- Best Presentation Paper Award
- 2018: OSDI travel grant
- 2015: ACM Student Research Competition 2nd Runner Award